

Foreword

Do You Want to Make a “Transformational Impact”? Michael Develin Shares his Advice and his Decades of Academic and Employment Experience

Dr. Michael Develin, Class of 1996, currently works as the Head of Data Science, Merchant Services at Shopify and is best known for his work as a mathematician in combinatorics and discrete geometry. He served as captain of the Math Team while at Stuyvesant but at age 16 entered Harvard University, where he was honored as a Putnam Fellow in the 1997 and 1998 competitions. After just three years, Develin earned his Ph.D. from University of California, Berkeley at age 22, writing his dissertation on Topics in Discrete Geometry under his advisor Bernd Sturmfels. Develin then conducted math research as the recipient of the 2003 American Institute of Mathematics Five-Year Fellowship. Over the course of his academic career, he published 31 research papers in mathematics journals, in fields such as combinatorics, discrete geometry, graph theory, number theory, and topology. Develin’s prior employment experience includes quantitative analysis, data science, and analytics at companies such as D.E. Shaw, Facebook, Instagram, and Remitly.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

How Stuyvesant Felt Like Home

I graduated from Stuyvesant in 1996, which is 26 years ago. I had always been interested in mathematics, but it became a reality for me at Stuyvesant. Much of my experience with math involved Math Team. More than a hundred students showed up to work on math problems every day, and realizing that there were not only a few people, but a lot of people pursuing this interest together was eye-opening. I really enjoyed Math Team and the bonding in the culture of math.

Outside of Math Team, I had the rare experience of not only taking higher level math classes, but the option to explore other fields of interest. I remember taking a writing workshop class, which I found much more compelling than reading Shakespeare for seven semesters in a row. But in the end, so much of my Stuy experience came down to the people. Being around others who are nerdy in the way that I was nerdy felt like home. I don’t think it would have felt that way at most other high schools.

From the Abstract World to the Real World

I went into graduate school, not because I had an explicit career goal, but because I purely loved working with numbers. While I enjoyed the experience of solving interesting and challenging problems, eventually I realized that working on abstract problems was not the trajectory for the rest of my life. Instead, I was looking for a direct, tangible impact I could have on society.

After I finished my post-doctoral research, I started working at a hedge fund, which is where I learned how to solve problems that were grounded in something concrete. By the time you graduate from college, you have been exposed to a lot of problems that have a right answer, but most of those problems are not connected to the real world. So in those corny word problems, like calculating the time the train leaves, it doesn't matter that it's a train. It's just a situation that you then turn into a math problem. So we emerge from school with that type of mindset and find that the real world does not actually work that way. In the work world, the process is much different; you try to get closer to the truth or build algorithms slightly better than the one before. I grew and learned more of that process after I started working at Facebook and Instagram.

These days, all the doors are open to you if you are fluent in numbers. All the industries have become and are becoming data-driven over time. Back when I was at Stuy, the sports industry was not using analytics like it is today. Now, the need everywhere is for people who can think in a structured way about problems and figure out the objective version of any situation.

What people often do not see is how data is used in tech companies: to give the user a voice. At the end of the day, it really came down to the question of how I can use data, which represents people's voices, to make better decisions.

A Generalist “Aiming for the Stars”

In retrospect, I'm most happy about the fact that I did not take just math classes during high school and college. I think what sets apart people who make an exceptional impact is the ability to excel in areas other than your core competency. There are some roles where you do need to be as good as possible at just math; being a professor is one of those roles. However, don't be afraid to diversify; don't feel like you need to become the world's best programmer. For example, the most impactful software engineers are not just programmers. They are the ones who can also think about the product and the customer, figure out how to architect things, or understand the business problem that leads to founding a company. But to make a truly transformational impact, it is important to develop general communication and comprehension skills and to widen your horizons. You are limiting yourself if you try to become really good at only one thing. When I see people top out in their careers, it's usually not because they fall short as a programmer or analyst, but it's because they can't put the bigger context around the technology, and they can't have impact outside of just the code that they write.

If you are aiming for the stars, it's important to be a bit of a generalist. In that sense, the core skill that I took away from the years of doing math is the ability to set up and frame a problem and figure out the logical problem-solving. You learn some of that in math and computer science, but understanding the broader world is also helpful to solve problems that we actually encounter.

So keep your ears open. Let's say you take a software engineering internship at some point during college. Try to speak not just with the software engineers; pay attention to what others say as well. Those may be the perspectives you will eventually need to understand to potentially make an impact on the world.

Embrace the Imposter Syndrome

It's important to try to learn about things you don't immediately understand. I think a fair number of data folks stay in their comfort zone with data, so it is very easy to just gravitate back to the analysis that's already familiar. And in the meantime, it always feels unfortunate when some writer folks describe the data folks as poor writers.

I'm not a professional caliber writer, but I've practiced a lot to become a better communicator. I'm never going to be the person who can talk to someone and immediately understand them and their life story. But even though I knew I wasn't going to be great at that, I have practiced talking to customers about their problems. It took time to take in their perspectives, understand them thoroughly myself, and then utilize that knowledge to address their problems.

The same process can apply to folks at Stuyvesant. Stuy students have been largely high achievers their entire lives. When you enter a new environment, whether it's high school, college, or a first job, you have the anxiety of being surrounded by really smart people. All of a sudden, you feel that you don't belong there, or you're out of your league. But I think it's important to embrace that Imposter Syndrome. Everyone encounters it to some degree, and there will always be problems you cannot solve. Don't pigeonhole yourself; be willing to embrace it and say, "I should try to improve in this area where I might currently not be that great at right now."

Keep your head up and try to explore; there's so much out there!

Mike Develin
November 2021